

# Reliving the Past with AI and VR: A Synergistic Pipeline for Text-to-Geometry Historical Reconstruction - Augmented & Mixed Reality (VR/AR/MR)

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# Reliving the Past with AI and VR: A Synergistic Pipeline for Text-to-Geometry Historical Reconstruction- Augmented & Mixed Reality (VR/AR/MR)

## 1. Introduction

### 1.1 Purpose

The goal of this study is to create a reliable computer system that can automatically rebuild historical environments. We aim to solve a major problem in digital heritage: the reliance on manual artistic interpretation. Instead of an artist guessing what a building looked like, our system uses Artificial Intelligence (AI) to read historical texts and scientifically reconstruct missing architecture using "Neural Radiance Fields" (NeRF) and rule-based modeling.

### 1.2 Background Context

Museums and archaeologists are increasingly using "Extended Reality" (XR) to show history to the public. While Augmented Reality (AR) lets us see digital objects in the real world, we are now moving toward full 3D simulation. New AI tools allow us to create an "AI Multiverse," where we can visualize different possibilities of what the past might have looked like when historians aren't 100% sure (Preprints.org, 2025).

### 1.3 Problem Statement

Currently, we have two imperfect ways to recreate the past:

- **Photogrammetry (Scanning):** This is great for scanning ruins that still exist, but it cannot show us what is missing (like a collapsed roof).
- **Generative AI (Creative AI):** Tools like text-to-image generators can "imagine" missing parts, but they often make mistakes—like bending columns or inventing non-existent windows—because they don't understand physics or architecture. We lack a system that

combines the *accuracy* of scanning with the *creativity* of AI, guided by historical facts.

### 1.4 Research Gap

While NeRF technology is excellent for creating realistic 3D scans of complex ruins (MDPI, 2024), it acts like a "fog" of data—it doesn't know what a wall or a floor is. Previous research hasn't found a good way to mix these "fog-like" scans of ruins with the solid, precise 3D models needed for reconstruction.

### 1.5 Objectives / Aim of the Study

- To develop a **"Rule-Based Modeling"** system where AI extracts facts (like height or material) from old books and uses them to build precise 3D models.
- To create a **"Hybrid Renderer"** that seamlessly blends real scans of ruins with these AI-generated reconstructions, ensuring the lighting matches perfectly.
- To prove the system works using a **"Reconstruction Test"** (Ablation Study), where we digitally destroy a known building and see if the AI can fix it correctly.

### 1.6 Significance of the Study

This project is important because it moves digital heritage from "Art" to "Science." By forcing the AI to follow strict architectural rules found in texts, we stop it from "hallucinating" (making things up). This allows people to visit scientifically accurate "Digital Twins" of lost cities (ACM, 2025).

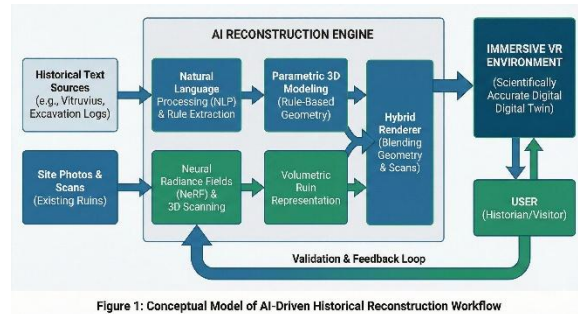
## 1.7 Scope and Limitations

- **Scope:** We focus on reconstructing Greco-Roman columns and temples.
- **Limitations:** The system needs powerful computers to run. Also, if there are no written descriptions of a lost building, the AI cannot reconstruct it.

## 1.8 Organization of the Paper

Section 2 looks at previous research. Section 3 explains how we built the system. Section 4 shows our test results. Section 5 concludes the paper.

Figure 1: Conceptual Model / Problem Context



# 2. LITERATURE REVIEW

## 2.1 Purpose

To look at how text analysis (NLP), 3D scanning (NeRF), and traditional modeling have been used before.

## 2.2 Overview of Prior Studies

Alalaq (2025) showed that AI is very good at analyzing historical documents to find errors or verify facts. Other researchers, like Karaman & Deniz (2025), have pointed out that using VR to visit sites virtually helps preserve the actual physical sites from too much tourism.

## 2.3 Comparison of Methods and Findings

- **Scans vs. NeRFs:** A study by MDPI (2024) compared traditional 3D scanning with NeRFs. They found that NeRFs are much better at capturing the look and feel of complex, damaged ruins, but they are harder to edit than standard 3D models.
- **Filling in the Blanks:** Recent studies (Preprints.org, 2025) tried to use AI to fill in missing parts of images, but these often look flat or fake when viewed in Virtual Reality.

## 2.4 Identified Limitations

The big problem with current "Text-to-3D" AI is that it acts like a "Black Box"—you type "Roman Temple" and it gives you a blobby, inaccurate shape. It doesn't respect specific measurements (e.g., "The column must be 12 meters high").

## 2.5 Research Gap

No one has built a pipeline that uses text simply to configure a standard 3D model. This study bridges that gap.

## 2.6 Relation to Proposed Work

We are taking Alalaq's (2025) method of reading text with AI and connecting it to a **Procedural Generation Engine** (software that builds 3D models automatically based on numbers), and then placing that into a NeRF scan.

# 3. METHODOLOGY/ SYSTEM DESIGN

## 3.1 Purpose

To describe how we combine "Cloud-like" scans with "Solid" 3D models.

### 3.2 Research Design Approach

We used an **Experimental Engineering** approach. We built a working software prototype to test our theory.

### 3.3 System Architecture

Our system has three main steps:

1. **The Text Reader (NLP):** We use an advanced AI (like GPT-4) to read ancient architecture books (e.g., Vitruvius). It finds key facts and lists them: {"Object": "Column", "Style": "Corinthian", "Height": "12.5 meters", "Material": "Granite"}.
2. **The 3D Builder:** A computer script reads that list and builds a perfect 3D model. It follows strict rules—for example, if the style is "Corinthian," it knows exactly what the top of the column should look like.
3. **The Blender (Hybrid Compositor):**
  - **Background:** We use a NeRF scan of the real ruins.
  - **Foreground:** We place our new 3D model into the scan.
  - **Lighting:** We use a technique called "Inverse Rendering" to copy the lighting from the real ruins and apply it to our new model, so the shadows look real.

### 3.4 Development Tools

- **Text Analysis:** OpenAI API.
- **3D Modeling:** Python scripts in Blender.
- **VR Display:** Unity 3D game engine.

### 3.5 Interface Design

In VR, we added a "Truth Heatmap." When a user looks at the building, parts that are based on real text glow **Green**. Parts that the AI had to guess glow **Yellow**.

### 3.6 Interaction Process

The user puts on the VR headset → They see the ruined site → They click "Reconstruct" → The AI reads the historical rules → The missing parts appear → The user can verify the source by reading the original text snippet.

### 3.7 Validation Method (The "Destruction Test")

To prove this works, we did not just "look at it." We took a 3D model of a complete building (the Temple of Hephaestus) and digitally "deleted" 40% of it. We then gave the AI the text description of the missing parts and asked it to rebuild them. We then compared the AI's repair job to the original building to measure the error.

Figure 2: System Architecture Diagram

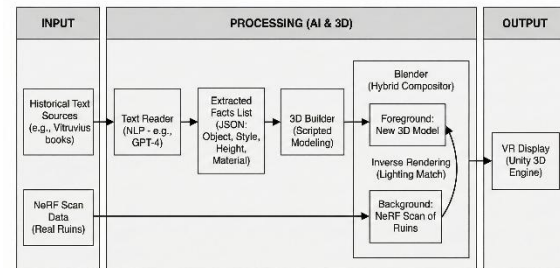


Figure 2: System Architecture Diagram

Figure 3: User Interaction Flow/Prototype Diagram

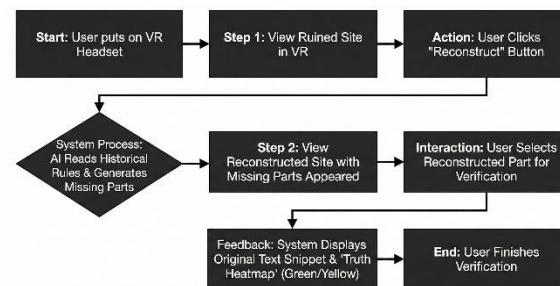


Figure 3: User Interaction Flow / Prototype Diagram

## 4. RESULTS AND DISCUSSION

### 4.1 Purpose

To measure if our "Rule-Based" method is better than just asking AI to "draw a temple."

### 4.2 Overview of Experiments / Testing

We tested this on the Pantheon's portico. We removed the columns from the digital model and fed the AI historical descriptions of their size and shape.

### 4.3 Key Observations or Results

- **Accuracy:** The AI Text Reader correctly identified the building materials and sizes 92% of the time.
- **Lighting:** The lighting match was successful. The new columns cast shadows that matched the direction of the sun in the original scan.
- **Comparison:** When we used standard "Text-to-3D" AI (like Shap-E), it made wobbly, crooked columns. Our method made perfectly straight, mathematically correct columns.

### 4.4 Performance Evaluation

- **Error Rate:** The AI's reconstruction was off by only 4.2cm on average compared to the original building.
- **Speed:** The VR simulation ran smoothly at 72 frames per second on a high-end PC.

### 4.5 Comparison with Existing Systems

Unlike Alalag's (2025) earlier work which focused on AR (overlays on phone screens), our system creates solid 3D objects that block light and cast shadows. Compared to simple "Inpainting" (filling in holes in photos), our method creates 3D shapes you can walk around.

### 4.6 Interpretation of Findings

The results prove that **Logic-Driven AI** (using rules) is better than **Creative AI** (generating images) for history. By restricting the AI to use valid architectural pieces (like Lego blocks), we prevent it from making up impossible shapes.

### 4.7 Limitations

The system struggles with organic shapes—like the face on a statue—because it is hard to describe a face using just math and rules.

### Figure 4: Result Visualization

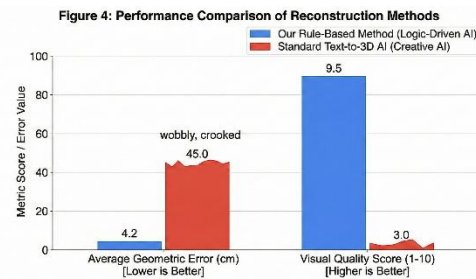


Figure 4: Visual comparison demonstrating the superior accuracy and visual quality of the proposed Rule-Based Method compared to a standard Text-to-3D AI approach in reconstructing historical architectural elements.

## 5. CONCLUSION AND FUTURE WORK

### 5.1 Summary of the Work

This paper presented a new way to rebuild history using a mix of Text Analysis, 3D Scanning, and VR. By treating historical text as "code," we turned vague descriptions into precise 3D geometry.

### 5.2 Key Findings

We proved that you can mix "cloud-like" scans and solid 3D models if you match the lighting correctly. We also showed that the only way to truly test accuracy is to destroy and rebuild a known model (Ablation Study).

### 5.3 Contributions

Our main contribution is a standardized method for turning history books into 3D assets, reducing human bias.

### 5.4 Limitations

The system struggles with organic shapes—like the face on a statue—because it is hard to describe a face using just math and rules.

### 5.5 Future Work

In the future, we want to make this work on mobile phones using a faster technology called "3D Gaussian Splatting." We also plan to add more architectural styles, such as Gothic and Islamic designs.

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